Artificial Intelligence

BS (CS) _SP_2026

Lab_01 Tasks



Learning Objectives:

1. Python Basic
2. Operators
3. Variables
4. Data Type
5. Control Statements
6. List, Sets, Tuples, Dictionaries
7. Classes and Objects
8. Local and Global Variables
9. Oop Concepts (Polymorphism, Abstraction, Inheritance, Encapsulation)

Lab Tasks

Submission Instructions

1. Create a separate file for each question with name *ROLLNO_SEC_LAB01* **e.g. i23XXXX_A_LAB01**
2. Compress all your **.py** files along with output **screenshots** directly into a **.zip file**.
3. Now you have to submit this zipped file on Google Classroom under your section heading.
4. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Q1. Write a Python script that defines a dictionary representing the inventory of a store.

Q2. Write a Python function that takes a list of numbers as input and returns the count of occurrences for each unique number in the list.

Input Example:

[1, 2, 3, 2, 1, 4, 3]

Expected Output:

1: 2

2: 2

3: 2

4: 1

Q3. Write a function that compares your and your friend's information, including name, age, and list of subjects. The function should:

- Identify the subjects that you are studying but your friend is not.
- Determine whether you are older than your friend.

Input Example:

Your Info:

{

"name": "xyz",

"age": 23,

"subjects": ["oop", "dsa", "pf", "os", "automata", "compiler", "web"] }

Friend's Info:

{

"name": "abc",
"age": 20,

"subjects": ["oop", "dsa", "compiler", "web"]

}

Expected Output:

Subjects you are studying but your friend is not:

pf

os

automata

You are elder than your friend.

Q4. Write a program that squares each value in a given list, then checks if the squared value is even or odd. Print the result in a readable format.

Input Example:

[1, 2, 3, 4]

Expected Output:

1: odd

4: even

9: odd

16: even

Q5. Create a base class Vehicle and derived classes Car and Bike.

- Create a base class Vehicle with:
  - **Attributes** : make (manufacturer), model, and year.
  - **Method** : display_info() that prints the details of the vehicle.

• Create a derived class Car that inherits from Vehicle:

▪ **Attributes** : Add an attribute num_doors (number of doors).

▪ **Method** : Override the display_info() method to include num_doors. • Create another derived class Bike that inherits from Vehicle:

▪ **Attributes** : Add an attribute type_of_bike (e.g., "mountain", "road"). ▪ **Method** : Override the display_info() method to include type_of_bike.

Create objects for both Car and Bike, set their attributes, and call the display_info() method to display the information.

Expected Output:

Car Details:

Make: Toyota

Model: Corolla

Year: 2020

Number of Doors: 4

Bike Details:

Make: Trek

Model: Marlin 7

Year: 2021

Type of Bike: Mountain

Q6. Create a Python class for a rectangle that can calculate and return both the area and diagonal of the rectangle.

$$Area = length * width$$

$$Diagonal = \sqrt{(length^2 + width^2)}$$

• Create an object of the Rectangle class with specific length and width • Call the area()and diagonal() methods, and print the results.

Input Example:

Length = 5
Width = 3

Expected Output:

Length: 5

Width: 3

Area: 15

Diagonal: 5.83

Q7. Write a Python program for a library management system using class inheritance. Design a base class **'LibraryItem'** with attributes like '**item_id**','**title**', and **'availability'**. Implement methods, including **'check_out()'** and **'return_item()'**, to manage borrowing and returning items. Derive two classes, **'Book'** and **'ResearchThesis'**, inheriting from **'LibraryItem'**. Include specific attributes (e.g.,**'author'** for books, **'Thesis_Topic'** for ResearchThesis) and customize the 'get_info()' method in each derived class for additional information. Implement methods such as 'calculate_penalty()' for late returns. To simulate library operations, create a function, 'simulate_library_operations(library_items)', that takes a list of item instances (mix of books and ResearchThesis) and demonstrates operations like checking out and returning items, displaying information, and calculating penalties for late returns.

Note: Ensure to focus on the class inheritance, method overriding, and polymorphic behavior while designing this library management system.